Ramco Institute of Technology, Rajapalayam
Department of Mechanical Engineering
Academic Year 2020-2021 (Even Semester)
Innovative Practices(s) Description

Name of the Faculty	:	Mr.T.Selvasundar, AP/Mechanical, R.Prabhakaran, AP/Mechanical
Subject Code and Title	:	CE8395 Strength of Materials for Mechanical Engineers
Year and Branch	:	II –Mechanical Engineering
Topics Covered	:	Shear Force and Bending Moment Diagram
Website Link	:	https://beamguru.com/beam/
No.of Students participated	:	47
Date & Time	:	27.03.2021 & 11.00AM to 11.50AM

Demonstration of Shear Force and Bending Moment Diagram Using Simulation Softwrae (Beamguru)

OBJECTIVE: To compare the Analytical Calculation, Shear force and Bending Moment diagram to Simulation Software results.

BEAMGURU.COM is an online calculator that generates Bending Moment Diagrams (BMD) and Shear Force Diagrams (SFD) for any statically determinate beams.

The following diagrams can also generate using the simulation software,

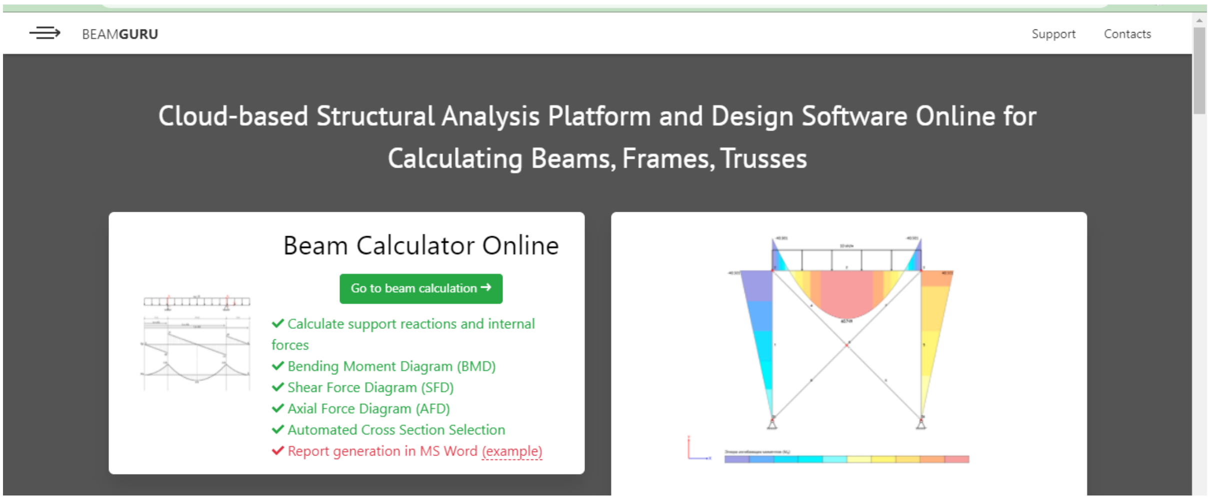
- Calculate support reactions and internal forces
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Axial Force Diagram (AFD)
- Automated Cross Section Selection

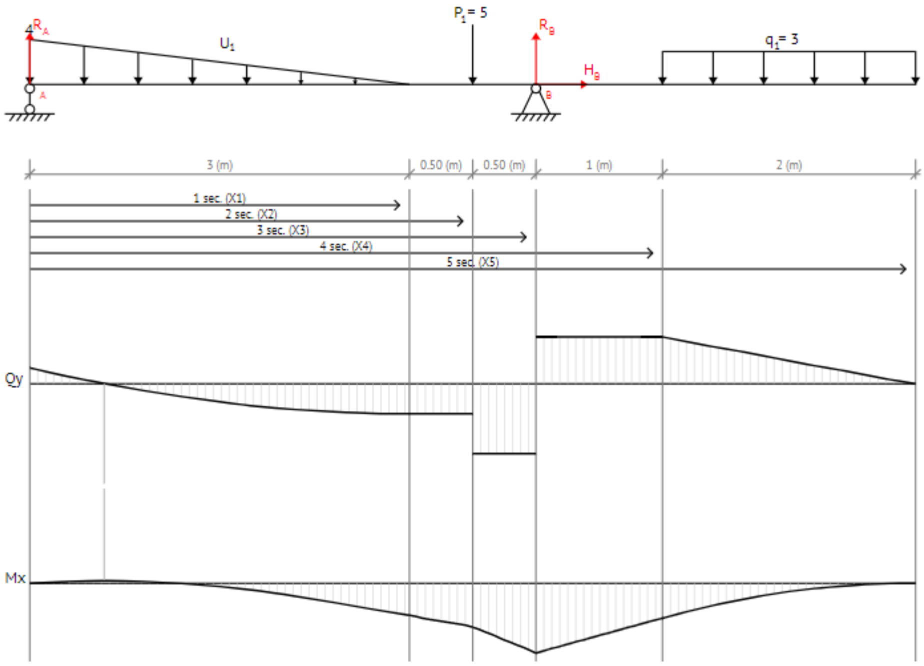
Shear force at any point along a loaded beam may be defined as the algebraic sum of all vertical forces acting on either side of the point on the beam.

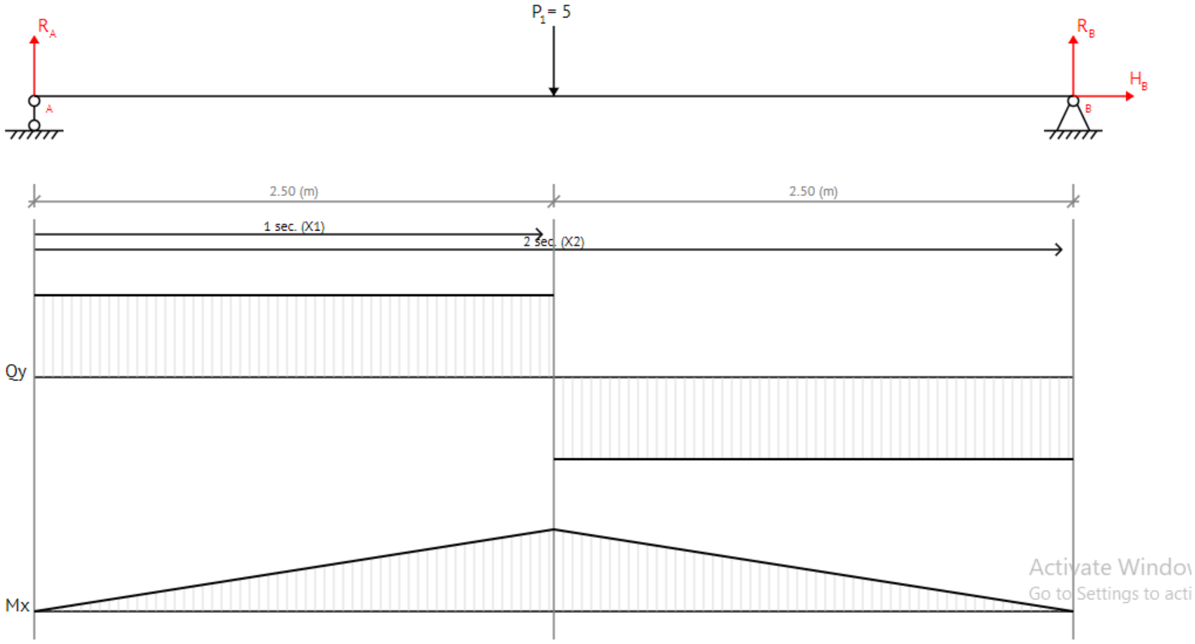
Bending moment at any point along a loaded beam may be defined as the sum of the moments due to all vertical forces acting on either side of the point on the beam.

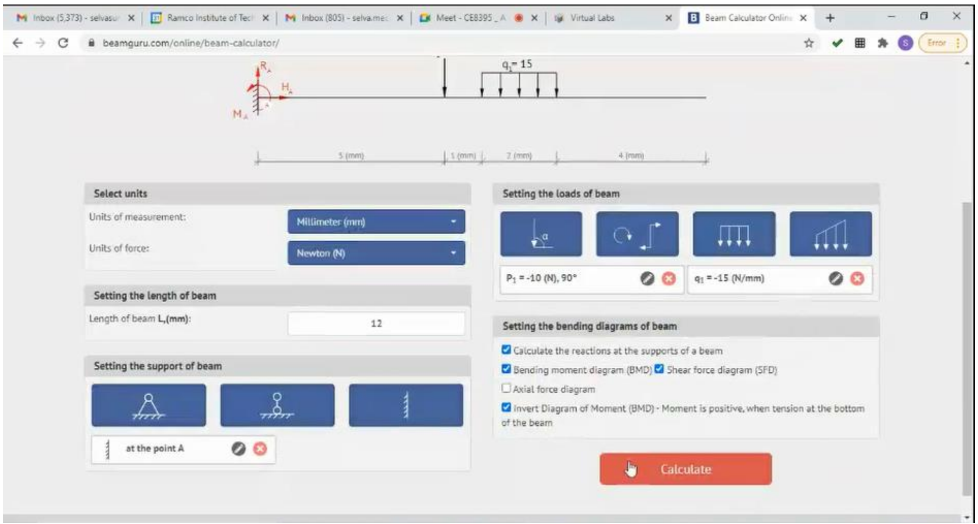
OUTCOME: The Cloud-based Structural Analysis Platform and Design Software Online for Calculating Beams help in attaining the following CO-PO mapping:

Course Outcome / Programme Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO12**
CO 2: The students will be able to mathematically predict the load transferring mechanism in beams and stress distribution due to shearing force and bending moment.	3	2	2	2	3	2	2	2






$P_1 = 5$
R_A
R_B
H_B
A
B
2.50 (m)
2.50 (m)
1 sec. (X1)
2 sec. (X2)
Qy
Mx
Activate Windows
Go to Settings to activate


beamguru.com/online/beam-calculator/
$q_1 = 15$
Select units
Units of measurement:
Millimeter (mm)
Units of force:
Newton (N)
Setting the length of beam
Length of beam L,(mm):
12
Setting the support of beam
at the point A
Setting the loads of beam
P_1 = -10 (N), 90°
q_1 = -15 (N/mm)
Setting the bending diagrams of beam
Calculate the reactions at the supports of a beam
Bending moment diagram (BMD)
Shear force diagram (SFD)
Axial force diagram
Invert Diagram of Moment (BMD) - Moment is positive, when tension at the bottom of the beam
Calculate